

L4: ML-Linear Regression - 3

Revision

- * Looked into car24 problem predicting price of used car
- * here we have target variable in training data, so its a supervised learning & target variable i.e. price can go from $0 \rightarrow \infty$ ~~then~~ so its a $\downarrow$ regression problem.
linear
- * Now as we look data some of the cols / features are in string we need them in ~~int~~ numbers so did the target encoding
- * Now ~~to~~ as the different features have different distribut, we did scaling. to understand features better, min-max scaling

* divided data on X_{train} X_{test}

* Trained the model
from `sklearn.linear_model` import
Linear Regression

```
model = LinearRegression()  
model.fit (  $X_{train}$ ,  $y_{train}$  )
```

then checked coefficients

`model.coef_`

`model.intercept_`

* looked at `coef_`

which is large that feature is
important for model.

larger absolute value of feature more
important feature is.

* ~~How~~ How to evaluate model = R^2

$$R^2 = 1 - \frac{RSS}{TSS}$$

$$RSS = \sum (\hat{y} - y)^2$$

$$TSS = \sum (\hat{y}_{avg} - y)^2$$

practical $R^2 = \{0, 1\}$

$R^2 = 0 = \text{avg. model}$

→ How fit method works
it uses gradient descent internally

$$w = w - \eta \frac{\partial L}{\partial w}$$

$$w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

what is ?

~~L = MSE~~

Suppose $\hat{y} = \omega^T x + \omega_0$

$$\hat{y} = \omega_1 x + \omega_0$$

$$\omega^T = [\omega_1]$$

$$L = \sum \frac{(y - \hat{y})^2}{N} = \frac{1}{N} \sum (y - (\omega_1 x + \omega_0))^2$$

$$\frac{\partial L}{\partial \omega_1} = \frac{1}{N} \sum 2 (y - (\omega_1 x + \omega_0)) \cdot (-x)$$

$$\frac{\partial L}{\partial \omega_0} = \frac{1}{N} \sum 2 (y - (\omega_1 x + \omega_0)) \cdot (-1)$$

$$\frac{\partial L}{\partial \omega_1} = \frac{1}{N} \sum 2 (y - \hat{y}) \cdot (-x)$$

$$\frac{\partial L}{\partial \omega_0} = \frac{1}{N} \sum 2 (y - \hat{y}) \cdot (-1)$$

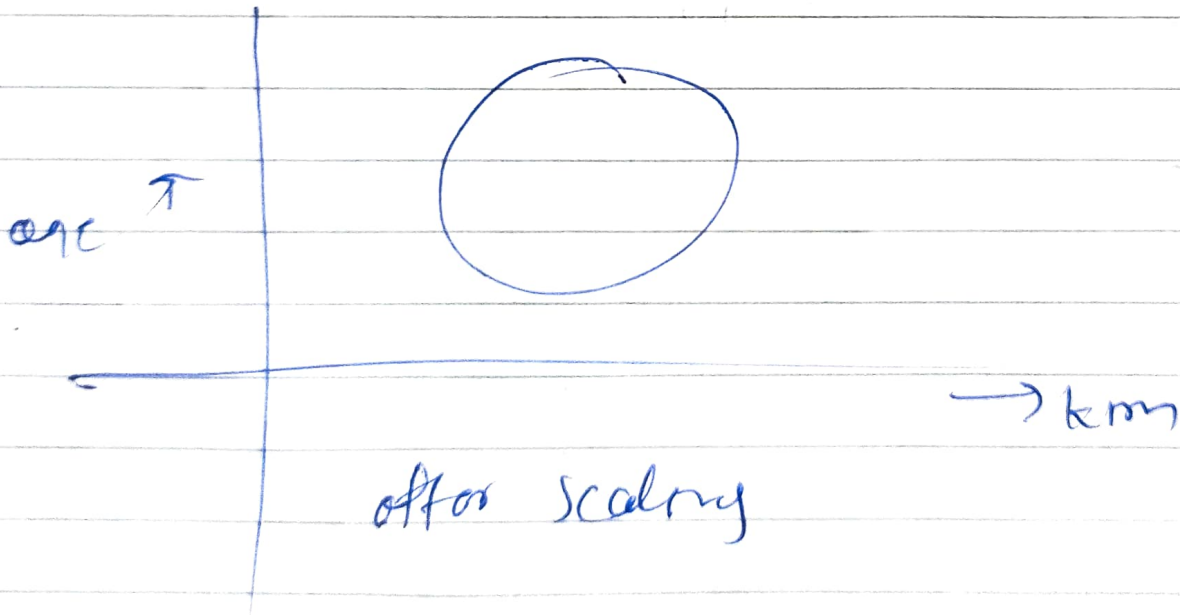
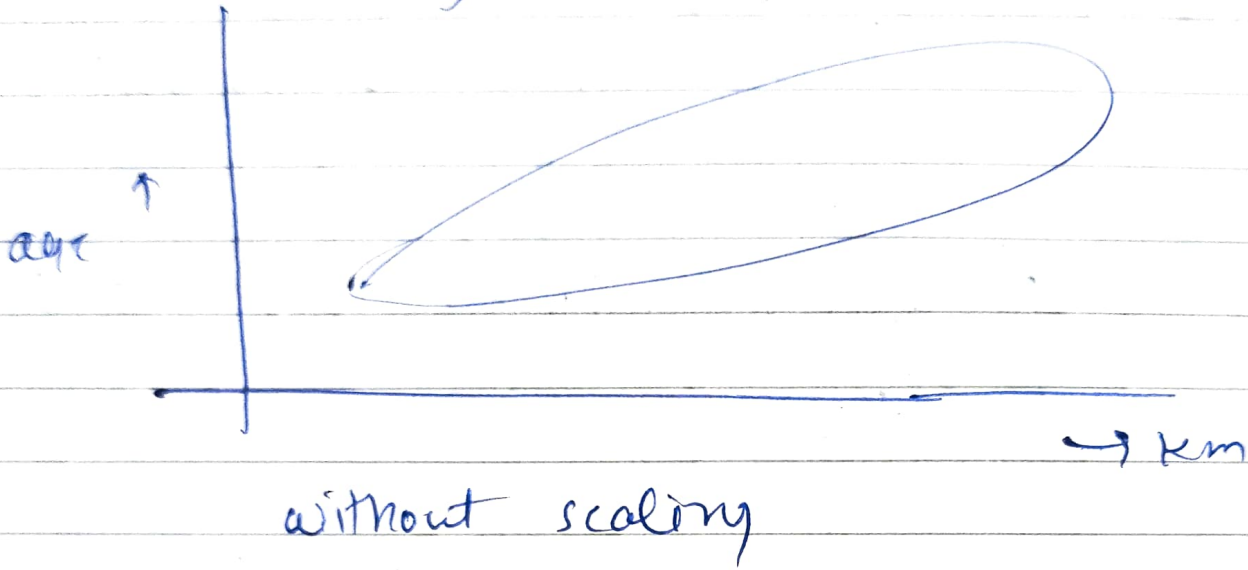
Look python code
named "L3-M2-LR2 - Linear-Regression
-from-scratch.ipynb"

The code in above .ipynb is also in
L3-M2-LR2-Notes.ipynb & in this
file this LR is also used in example
Look this file only

Upto this point everything is
covered in file

"L3-M2-LR2-Notes.ipynb"

Why scaling → basically to have all feature values in same range



When the model do not give good result. 2 ~~the~~ problems may be

- ① data
- ② algorithm (model $\rightarrow$ e.g. linear regressⁿ, etc)

data problem can be solved by
scaling $\rightarrow$ standardization / normalization
need to check what
are these.

algorithm problem

Need to check whether linear
regressⁿ works or not etc.

Adjusted R^2

Consider following data

X	Y	after applying model/ (CLR model) R^2
Sq.m	price	
100	2CR	= 0.11
700	4CR	
400	4CR	
200	2CR	

which means the model is 11% better than mean model

If model correct then it means X explains only 11% of price.

Now we add another feature

X1	X2	price
local	sq. m	
Mumbai	100	200
Solapur	700	400
Solapur Mumbai	400	400
Mumbai Solapur	200	220

Now model will give $R^2 = 0.9$
 90% of price can be explained
 X1 & X2

If we add another feature
 will R^2 be $< 90\%$.

NO.

~~However~~ R^2 may $> 90\%$ but ~~not~~ not
 $< 90\%$.

suppose add new feature serial
 number, which will not affect
 the price, here model training
 will serial number weight as 0.

if feature is imp. $R^2 \uparrow$ if not
then R^2 remains constant.

if feature removed the R^2 may
go down.

$$\text{Adjusted } R^2 = 1 - \left[\frac{(1 - R^2) \overset{\text{rows}}{(n-1)}}{\underset{\substack{\uparrow \\ \text{feature/cat}}}{n-d-1}} \right]$$

why this formula came

suppose ① n features $R^2 = 0.85235$
② 2 " $R^2 = 0.85$

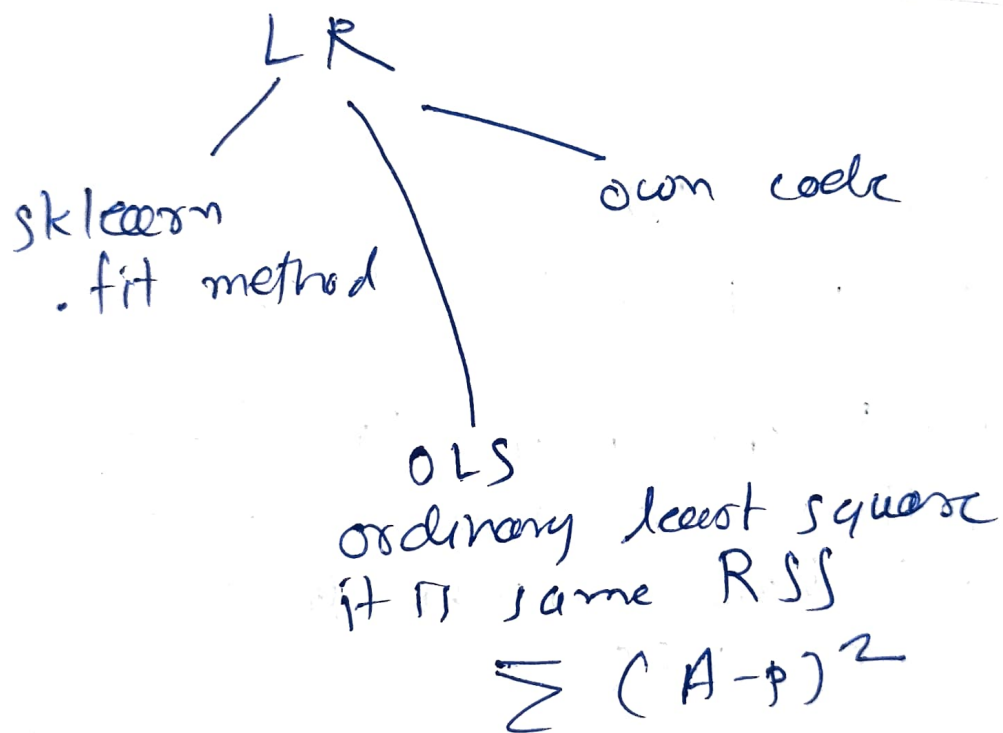
then we should choose model ②

So Adjusted R^2 takes account
of feature numbers

$X_{\text{train}} \rightarrow R^2 \rightarrow \text{Adj } R^2$

$X_{\text{test}} \rightarrow R^2 \rightarrow \text{Adj } R^2$

i) difference in R^2 & $\text{Adj } R^2$
then remove features & train again



After this, look at
L4 - ML + R3 - notes. i pymb
for why OLS is used

Recap

* Coding GD

* what happens if not done feature
scaling

* R^2 $\uparrow$ never $\downarrow$ if feature $\uparrow$

* Adj. R^2 penalizes if necessary if
not change much if feature $\uparrow$

* OLS = RSS

we get stats along with OLS

$p < 0.05 \rightarrow \text{var} \sim$

$p \neq 0.05 \rightarrow \text{var} \times$